

Computation Graphs, Backpropagation and Autodiff

How gradients flow through neural networks

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ES 667 · Deep Learning · IIT Gandhinagar

Backprop vocabulary

Lecture 2 built the forward map; today we reverse it

Last time, an MLP turned an input into a prediction and then one scalar loss. Today we reuse that **same graph** in reverse:

General pattern

Forward: compute values

$$x \rightarrow f_{\theta}(x) = \hat{y} \rightarrow \mathcal{L}$$

Backward: compute sensitivities loss
sensitivity for every parameter θ_j

Concrete scalar example

Forward: with $(w, x, b, y) = (2, 3, 1, 10)$

$$\hat{y} = 2 \cdot 3 + 1 = 7, \quad \mathcal{L} = (7 - 10)^2 = 9$$

Backward:

$$\left(\frac{\partial \mathcal{L}}{\partial w}, \frac{\partial \mathcal{L}}{\partial b} \right) = (-18, -6)$$

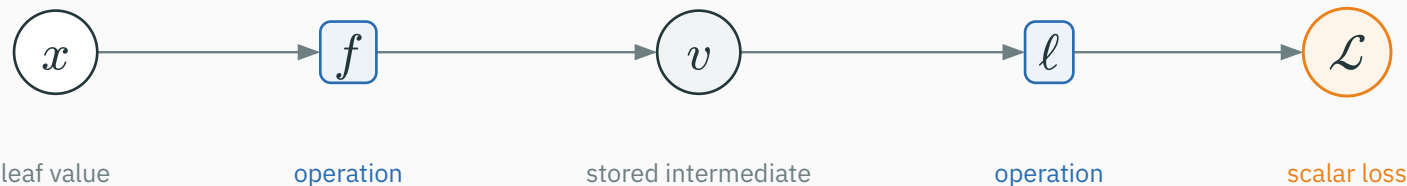
same executed graph · values flow forward, gradients flow backward

Read the notation before the algorithm

Symbol	Meaning in this lecture
u, v, z	scalars
$\mathbf{x}, \mathbf{z}, \boldsymbol{\theta}$	vectors (bold lowercase)
$\mathbf{W}, \mathbf{X}$	matrices (bold uppercase)
$\partial\mathcal{L}/\partial u$	how the loss changes when scalar u changes
$\partial\mathcal{L}/\partial\boldsymbol{\theta}$	all parameter derivatives, arranged like $\boldsymbol{\theta}$

We keep the full $\partial\mathcal{L}/\partial(\cdot)$ notation visible throughout: the numerator is always the final loss; the denominator names the stored value whose derivative we want.

Our computation-graph convention



white circle	given value: input, parameter, or target — a graph leaf
gray circle	computed intermediate value, saved for backward when needed
blue box	operation: consumes value(s) and produces a value
gray arrow	forward dependency from a value to the operation that uses it
orange circle	final scalar loss $\mathcal{L}$; seeds backward with $\partial \mathcal{L} / \partial \mathcal{L} = 1$

circles store values • boxes transform them • arrows show dependency

Symbolic = formula $\rightarrow$ formula. A person or a computer-algebra system may do the algebra.

$$\mathcal{L} = (wx + b - y)^2, \quad (w, x, b, y) = (2, 3, 1, 10).$$

Route	What it does	What it returns here
symbolic	derive a formula	$\frac{\partial \mathcal{L}}{\partial w} = 2(wx + b - y)x$; substitute $\rightarrow -18$
finite difference	evaluate $\mathcal{L}(w \pm \varepsilon)$	approximate number at $w = 2$: -18
reverse- mode AD	run local rules backward	number for this run: $(-6) \cdot 3 = -18$

formula · approximate value · reverse sweep through the executed graph

Reverse-mode AD computes the derivative from local rules, up to floating-point arithmetic. Central difference independently approximates the same value: $g_{\text{FD}} = \frac{\mathcal{L}(\theta_j + \varepsilon) - \mathcal{L}(\theta_j - \varepsilon)}{2\varepsilon}$.

```
import torch
def loss(w): return (w*3.0 + 1.0 - 10.0)**2
w, eps = 2.0, 1e-5
g_fd = (loss(w + eps) - loss(w - eps)) / (2*eps)
wt = torch.tensor(w, dtype=torch.float64, requires_grad=True)
loss(wt).backward()
g_ad = wt.grad.item()
print(g_fd, g_ad) # both approximately -18
```

Use float64 and check only a few coordinates. Finite differences debug gradients; they do not train models because every checked coordinate needs two extra forward passes.

Why train with reverse-mode autodiff?

Method	Exact?	Cost for P parameters	Use it for
symbolic	exact formula before numerical evaluation	expression-dependent; algebra can grow	deriving / reasoning
finite difference	approximate	$2P$ loss evaluations	checking
reverse mode	exact local rules; floating-point values	one reverse sweep for scalar $\mathcal{L}$	training

Symbolic to reason • finite differences to check • reverse mode to train.

Backprop returns every parameter sensitivity in one sweep

For each parameter coordinate, gradient descent uses

$$\theta_{j,t+1} = \theta_{j,t} - \eta \frac{\partial \mathcal{L}}{\partial \theta_j}.$$

η is the learning rate; the derivative supplies the direction and magnitude of the local change. A modern network has $P = 10^7$ to 10^{11} parameters.

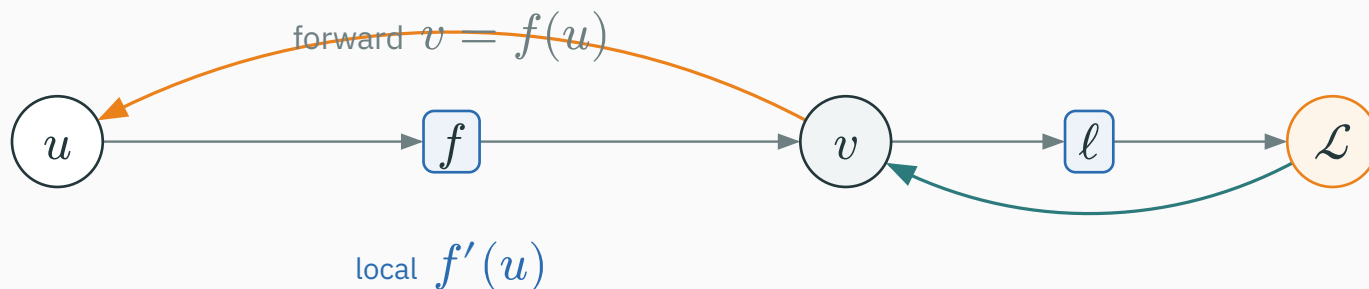
We need all P derivatives $\partial \mathcal{L} / \partial \theta_j$. Backprop computes them together in **one reverse sweep**, without perturbing parameters one at a time.

The route from one local operation to an entire network

Stage	Question we answer
1 · local rule	How does one operation pass a gradient backward?
2 · branches	Why do gradient contributions add?
3 · worked graph	Can we reproduce every derivative numerically?
4 · layers + MLP	How do scalar rules become vector and matrix formulas?
5 · depth + gradient flow	What happens when many local gradients multiply?

One local operation

Zoom in on one operation. The value circle u enters the boxed operation f ; its output is stored in the value circle v .

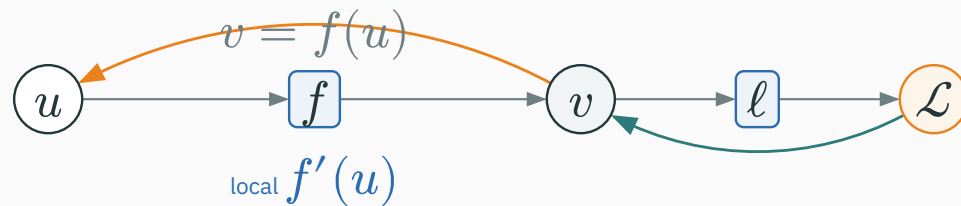


backward computes $\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot f'(u)$

Every operation multiplies the arriving gradient by its local gradient

D DERIVATION

For the operation $v = f(u)$, the chain rule is



$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot f'(u)$$

Downstream gradient

sent back to u

$$\frac{\partial \mathcal{L}}{\partial u}$$

Upstream gradient

arriving at v

$$\frac{\partial \mathcal{L}}{\partial v}$$

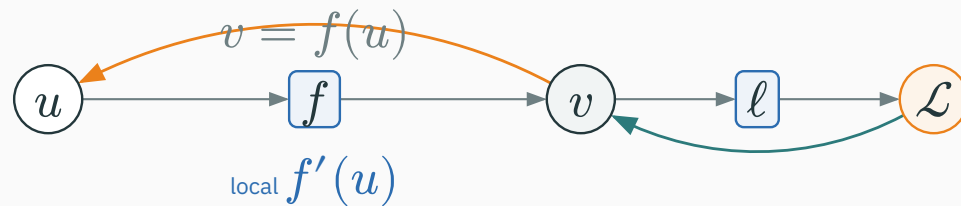
Local gradient

this operation's slope

$$f'(u) = \frac{\partial v}{\partial u}$$

The chain rule is one reusable local rule

For every scalar operation $v = f(u)$:

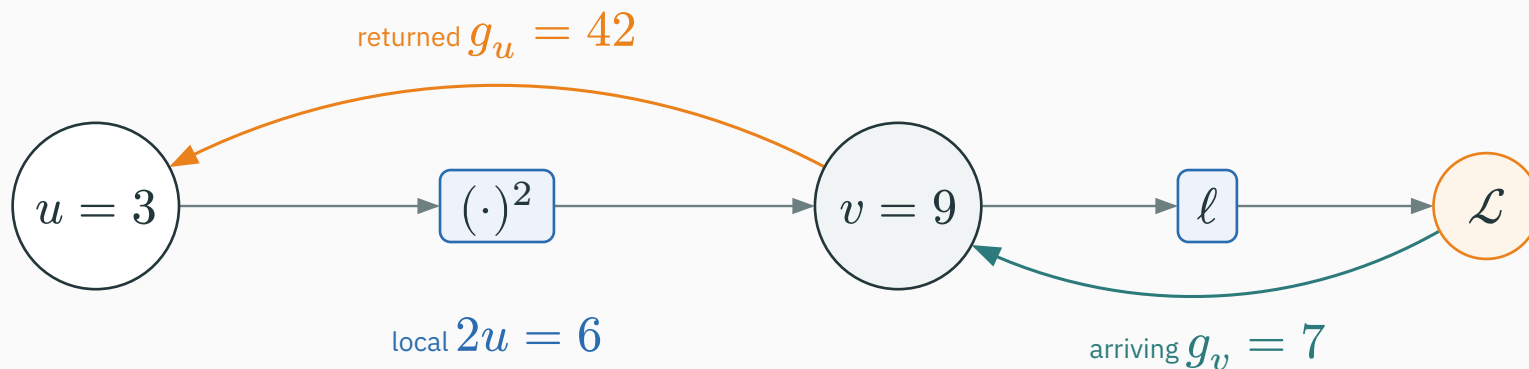


$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot \frac{\partial v}{\partial u}.$$

The operation multiplies its **upstream gradient** by its **local gradient** to produce the **downstream gradient**. It needs no global formula for the whole network.

Example: a square operation

For $v = u^2$ at $u = 3$, the forward value is $v = 9$ and the **local slope** is $\partial v / \partial u = 2u = 6$. Suppose the rest of the graph sends $g_v = \partial \mathcal{L} / \partial v = 7$ back to this square.

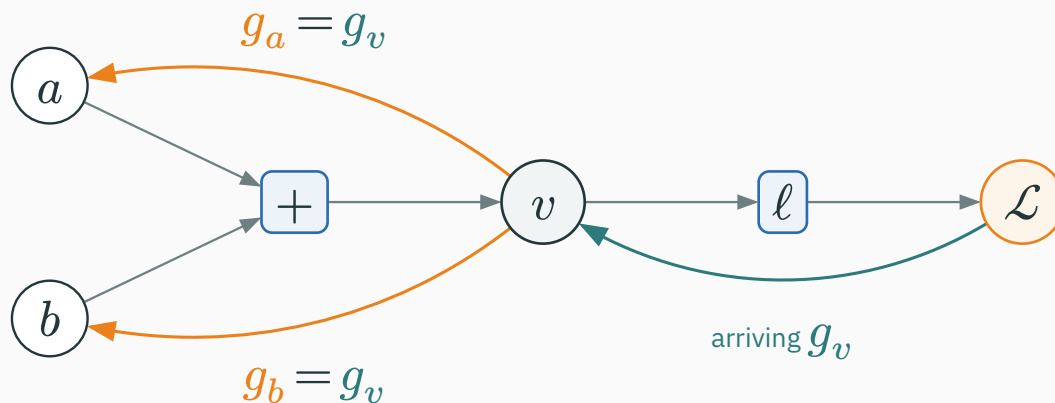


$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot 2u = 7 \cdot 6 = 42.$$

The square operation multiplies the arriving $\partial \mathcal{L} / \partial v = 7$ by its **local gradient** $2u = 6$ to produce $\partial \mathcal{L} / \partial u = 42$.

Addition copies the arriving gradient

Operation $v = a + b$. Let $g_v = \partial \mathcal{L} / \partial v$ arrive. Local derivatives: $\partial v / \partial a = 1$, $\partial v / \partial b = 1$.

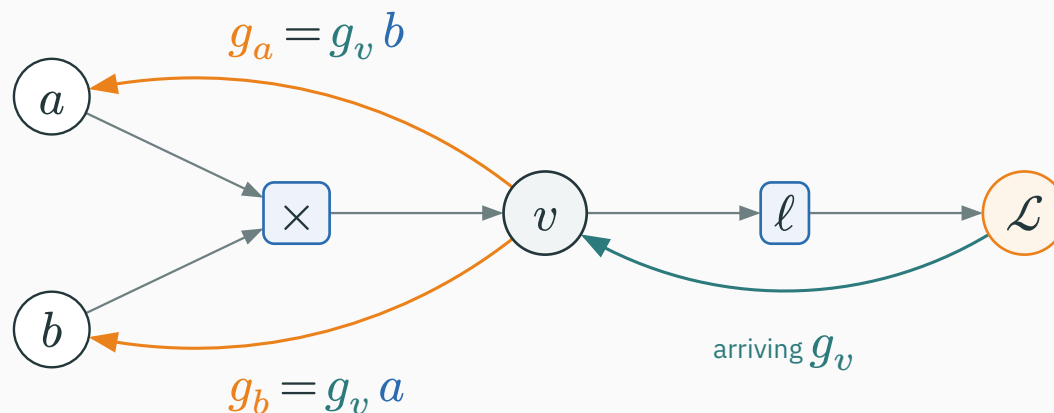


$$\frac{\partial \mathcal{L}}{\partial a} = \frac{\partial \mathcal{L}}{\partial v} \cdot 1, \quad \frac{\partial \mathcal{L}}{\partial b} = \frac{\partial \mathcal{L}}{\partial v} \cdot 1.$$

+ copies the arriving derivative into a downstream derivative for every input.

Multiplication scales by the other input

Operation $v = ab$. Let $g_v = \partial \mathcal{L} / \partial v$ arrive. Local derivatives: $\partial v / \partial a = b$, $\partial v / \partial b = a$.

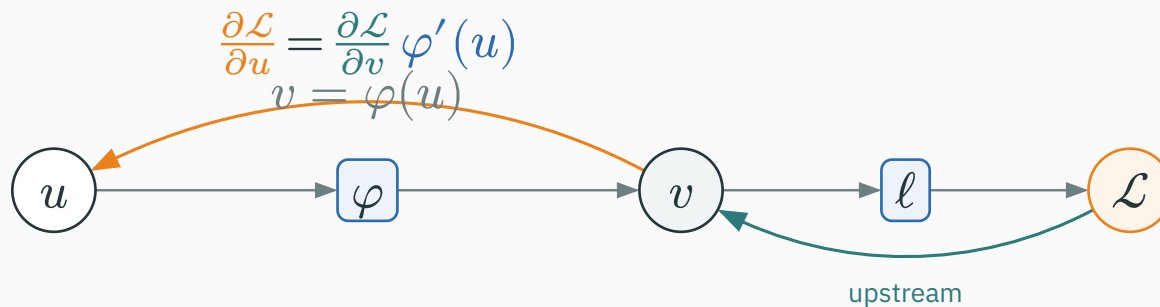


$$\frac{\partial \mathcal{L}}{\partial a} = \frac{\partial \mathcal{L}}{\partial v} \cdot b, \quad \frac{\partial \mathcal{L}}{\partial b} = \frac{\partial \mathcal{L}}{\partial v} \cdot a.$$

$\times$ uses the **other input as its local multiplier**.

| If $a = 2$, $b = 5$, and $\partial \mathcal{L} / \partial v = 3$, then $\partial \mathcal{L} / \partial a = 15$ and $\partial \mathcal{L} / \partial b = 6$.

Operation $v = \varphi(u)$ (elementwise nonlinearity). Local derivative: $\partial v / \partial u = \varphi'(u)$.



$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot \varphi'(u).$$

| **sigmoid** σ : $\varphi'(u) = \sigma(u)(1 - \sigma(u))$

| **ReLU**: $\varphi'(u) = 1$ for $u > 0$, and 0 for $u < 0$

Pop quiz: what crosses an inactive ReLU?

QUESTION

Let $v = \text{ReLU}(u)$. If $u = -2$ and the upstream gradient is $\partial \mathcal{L} / \partial v = 5$, what downstream gradient $\partial \mathcal{L} / \partial u$ is sent back?

A. 5

B. -10

C. 0

D. undefined because $u < 0$

Answer: an inactive ReLU blocks the gradient

A ANSWER

Correct: C — 0

Why: For $u < 0$, the local gradient is $\partial v / \partial u = 0$. Therefore downstream = upstream $\times$ local = $5 \times 0 = 0$.

Forward: $u = -2 \rightarrow v = 0$

Backward: $5 \times 0 \rightarrow \partial \mathcal{L} / \partial u = 0$

At exactly $u = 0$, ReLU's ordinary derivative is undefined. Autodiff software must choose a convention; commonly it returns 0.

The rule table — memorize this

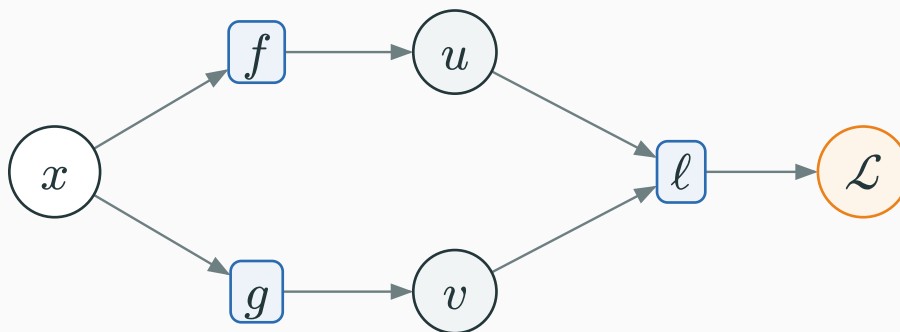
Forward operation	Contribution sent backward
$v = a + b$	$\frac{\partial \mathcal{L}}{\partial a} += \frac{\partial \mathcal{L}}{\partial v} \cdot 1$ $\frac{\partial \mathcal{L}}{\partial b} += \frac{\partial \mathcal{L}}{\partial v} \cdot 1$
$v = ab$	$\frac{\partial \mathcal{L}}{\partial a} += \frac{\partial \mathcal{L}}{\partial v} \cdot b$ $\frac{\partial \mathcal{L}}{\partial b} += \frac{\partial \mathcal{L}}{\partial v} \cdot a$
$v = u^2$	$\frac{\partial \mathcal{L}}{\partial u} += \frac{\partial \mathcal{L}}{\partial v} \cdot 2u$
$v = \varphi(u)$	$\frac{\partial \mathcal{L}}{\partial u} += \frac{\partial \mathcal{L}}{\partial v} \cdot \varphi'(u)$

Use +=, not = — a value consumed by several operations accumulates gradient from **all** of them.

Gradients accumulate at branches

Why gradients add at a branch

Imagine nudging x from 4 to 4.01. The **same nudge** travels down both routes to the loss:



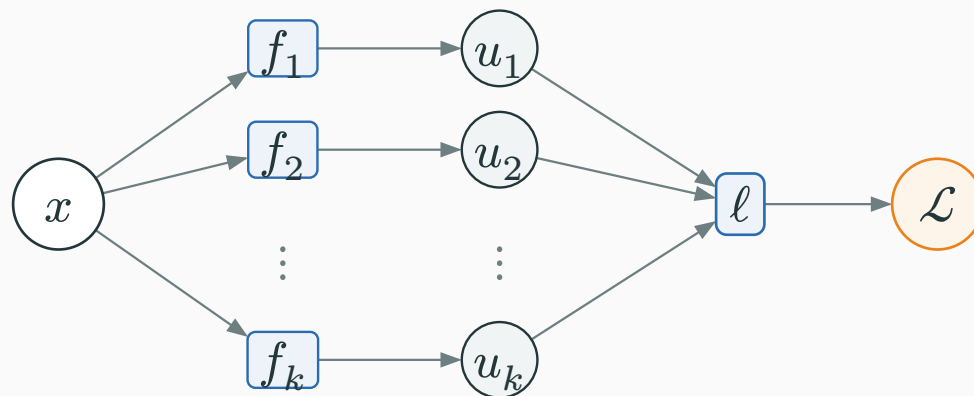
Via $u = x^2$: the local slope is $2x = 8$, so $\Delta x = 0.01$ changes the loss by about $8 \cdot 0.01 = 0.08$.

Via $v = 3x$: the local slope is 3, so the same nudge changes the loss by about $3 \cdot 0.01 = 0.03$.

$$\text{total slope} = \text{total loss change} \div \text{nudge} = \frac{0.08+0.03}{0.01} = 8 + 3 = 11$$

The multivariable chain rule adds every path

The nudge experiment generalizes. If x influences $\mathcal{L}$ through $u_1, \dots, u_k$:



branch i contributes upstream $\frac{\partial \mathcal{L}}{\partial u_i} \times$ local $\frac{\partial u_i}{\partial x}$

$$\frac{\partial \mathcal{L}}{\partial x} = \sum_{i=1}^k \frac{\partial \mathcal{L}}{\partial u_i} \frac{\partial u_i}{\partial x}.$$

each path contributes one chain-rule product; the products add

Worked branch example

Let $u = x^2$, $v = 3x$, $\mathcal{L} = u + v = x^2 + 3x$. By symbolic differentiation — derive the formula first:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x + 3, \quad \text{at } x = 4: \quad 2(4) + 3 = 11.$$

Now let us get the **same** answer by backprop.

Forward at $x = 4$: $u = 16$, $v = 12$, $\mathcal{L} = 28$. **Backward:** seed $\partial\mathcal{L}/\partial\mathcal{L} = 1$; the add operation gives $\partial\mathcal{L}/\partial u = 1$ and $\partial\mathcal{L}/\partial v = 1$.

$$\left(\frac{\partial\mathcal{L}}{\partial x}\right)_{\text{via } u} = 1 \cdot 2x = 8, \quad \left(\frac{\partial\mathcal{L}}{\partial x}\right)_{\text{via } v} = 1 \cdot 3 = 3.$$

$$\frac{\partial\mathcal{L}}{\partial x} = 8 + 3 = 11. \quad \checkmark$$

PyTorch adds the same two paths

```
import torch

x = torch.tensor(4.0, requires_grad=True)
u = x**2          # path 1
v = 3*x           # path 2
L = u + v
L.backward()

print(x.grad)      # tensor(11.)
```

| **via u :** $\partial_{\frac{u}{\partial}} x = 2x = 8$

| **via v :** $\partial_{\frac{v}{\partial}} x = 3$

autograd accumulates both downstream contributions: $8 + 3 = 11$

The central worked example

Predict $\hat{y} = wx + b$, squared-error loss $\mathcal{L} = (\hat{y} - y)^2$. Concrete numbers:

$x = 3, \quad w = 2, \quad b = 1, \quad y = 10.$

Compute $\partial\mathcal{L}/\partial w$, $\partial\mathcal{L}/\partial x$, $\partial\mathcal{L}/\partial b$, and $\partial\mathcal{L}/\partial y$ by backpropagation; verify the four values with PyTorch.

Break into primitives

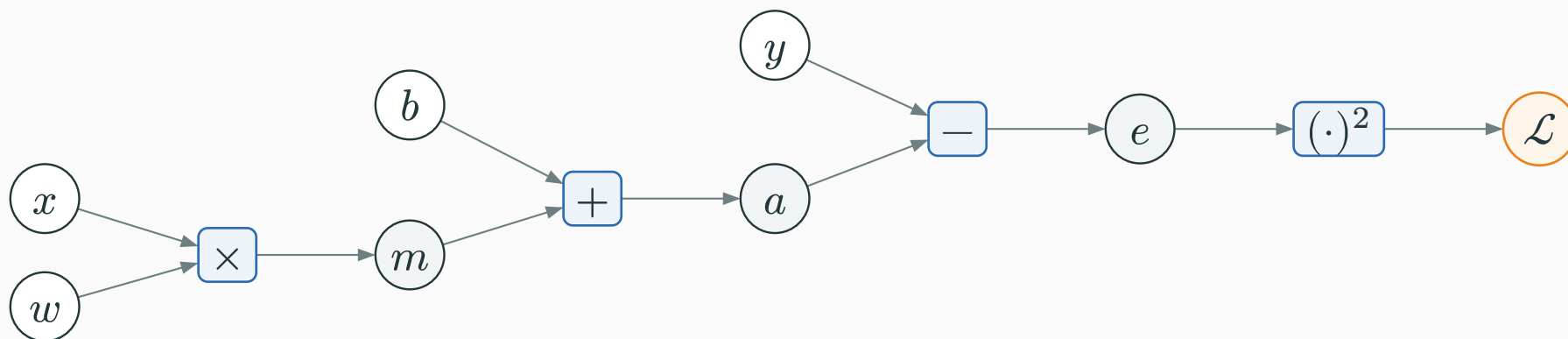
Decompose the loss into elementary operations and name each stored result:

$$m = wx, \quad a = m + b, \quad e = a - y, \quad \mathcal{L} = e^2.$$

value m after $\times$ $\cdot$ value a after $+$ $\cdot$ value e after $-$ $\cdot$ value $\mathcal{L}$ after $(\cdot)^2$

The symbols m , a , and e name stored intermediate values. They are circles in the graph; the four operations are boxes.

The computation graph

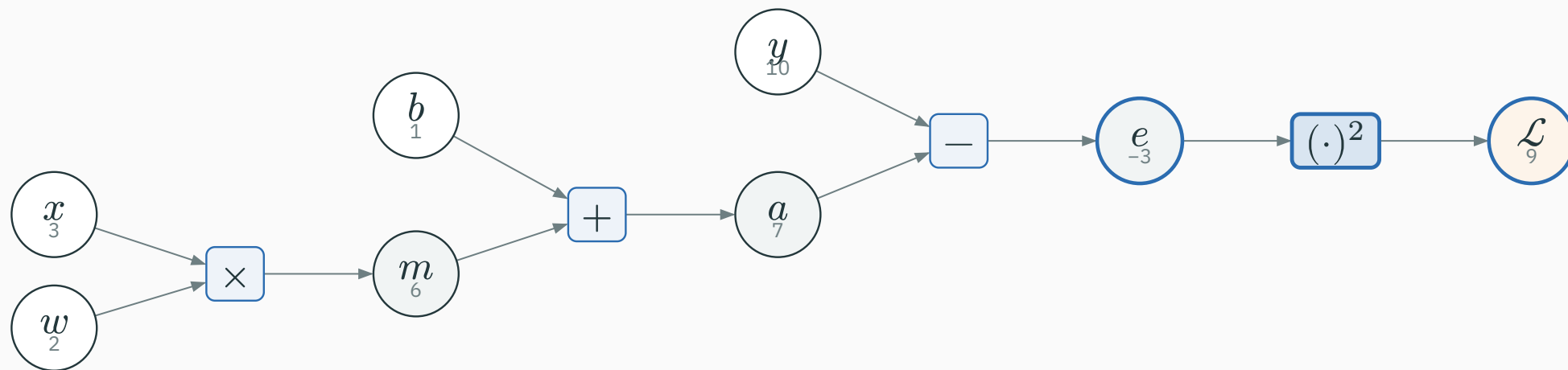


Given values	w, x, b, y	Stored values	m, a, e
Operations	$\times, +, -, (\cdot)^2$	Scalar loss	$\mathcal{L}$

Stored value	Operation that produces it	Numeric value
m	$w \times x$	$2 \cdot 3 = 6$
a	$m + b$	$6 + 1 = 7$
e	$a - y$	$7 - 10 = -3$
$\mathcal{L}$	e^2	$(-3)^2 = 9$

$\mathcal{L} = 9$ — now run the graph **backward**

Seed the output: $\partial \mathcal{L} / \partial \mathcal{L} = 1$.

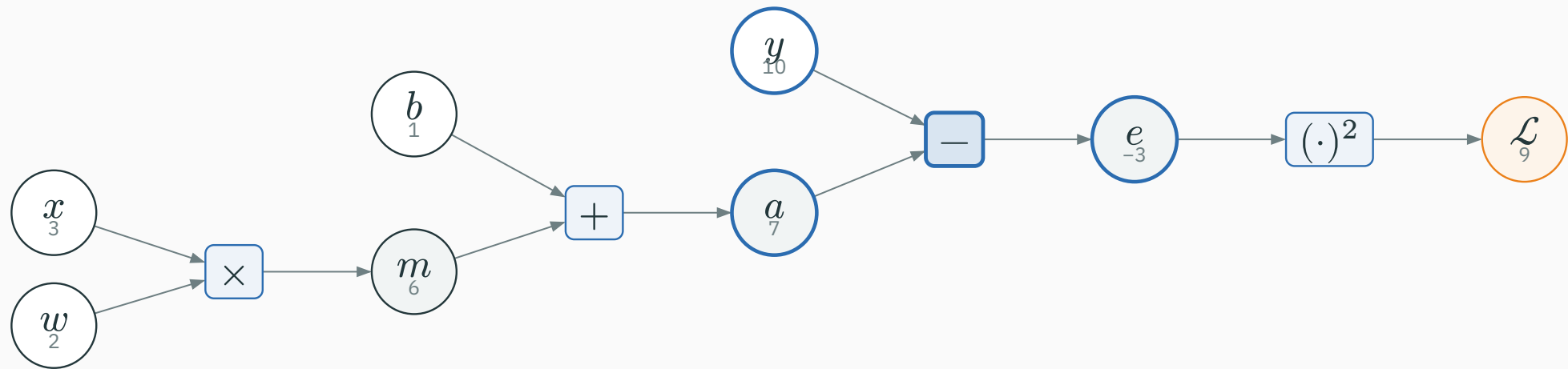


Square operation $\mathcal{L} = e^2$, local $\partial \mathcal{L} / \partial e = 2e$:

$$\frac{\partial \mathcal{L}}{\partial e} = 1 \cdot 2e = 1 \cdot 2(-3) = -6.$$

Backward: the subtraction operation

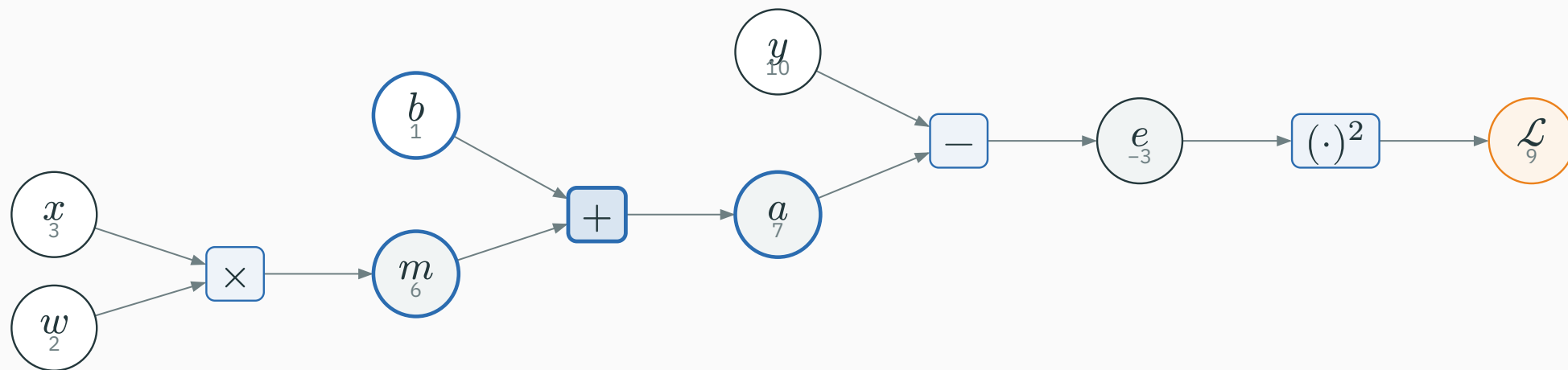
Subtraction operation $e = a - y$, locals $\partial e / \partial a = 1$, $\partial e / \partial y = -1$:



$$\frac{\partial \mathcal{L}}{\partial a} = \frac{\partial \mathcal{L}}{\partial e} \cdot 1 = -6 \quad \frac{\partial \mathcal{L}}{\partial y} = \frac{\partial \mathcal{L}}{\partial e} \cdot (-1) = 6$$

The target y also receives a derivative. We would use $\partial \mathcal{L} / \partial y$ if y were itself a learned quantity.

Addition operation $a = m + b$ **copies** its gradient:

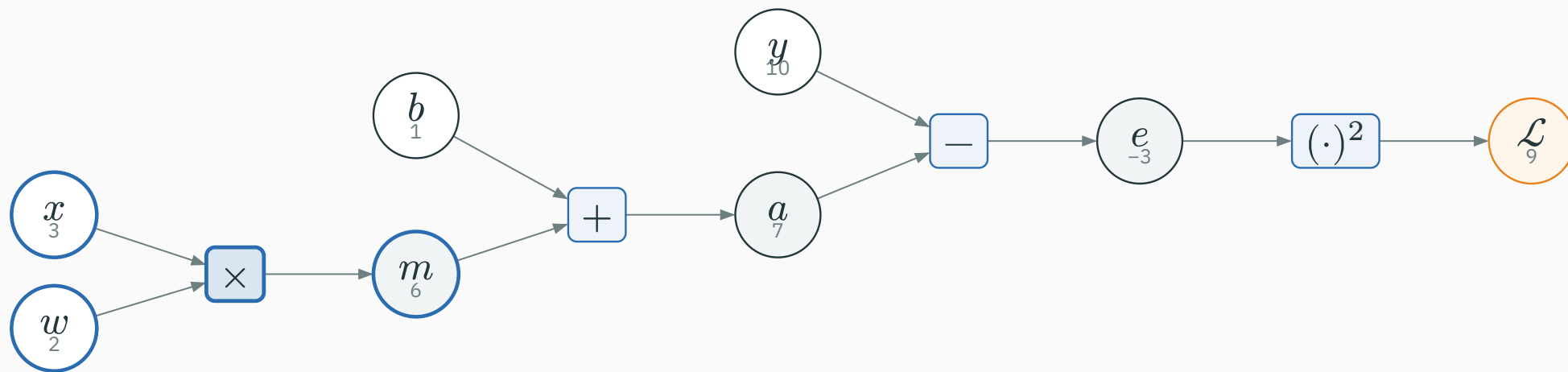


$$\frac{\partial \mathcal{L}}{\partial m} = \frac{\partial \mathcal{L}}{\partial a} = -6, \quad \frac{\partial \mathcal{L}}{\partial b} = -6.$$

So $\partial \mathcal{L} / \partial b = -6$ is one of our final answers.

Backward: the multiplication operation

Multiplication operation $m = wx$ **sends the other input:**



$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial m} \cdot x = -6 \cdot 3 = -18$$

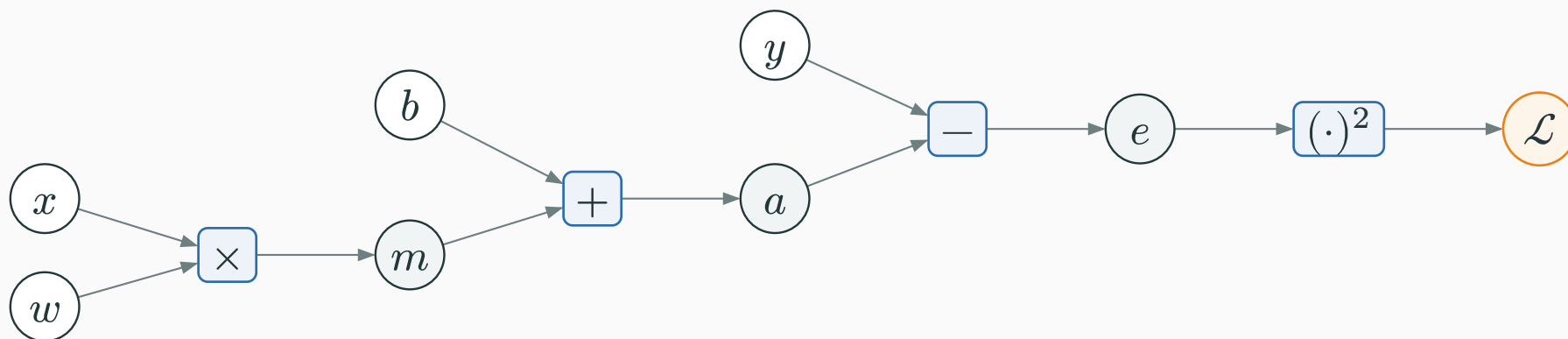
$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial m} \cdot w = -6 \cdot 2 = -12$$

$$\partial \mathcal{L} / \partial w = -18, \quad \partial \mathcal{L} / \partial x = -12$$

$\partial \mathcal{L} / \partial w$	$\partial \mathcal{L} / \partial b$	$\partial \mathcal{L} / \partial x$	$\partial \mathcal{L} / \partial y$
-18	-6	-12	6

One backward sweep produced **all four** partial derivatives — no formula re-derivation per parameter.

All stored values and their gradients



first half	w	x	m	b
forward value	2	3	6	1
gradient	$g_w = -18$	$g_x = -12$	$g_m = -6$	$g_b = -6$

second half	a	y	e	$\mathcal{L}$
forward value	7	10	-3	9
gradient	$g_a = -6$	$g_y = 6$	$g_e = -6$	$g_{\mathcal{L}} = 1$

Keep w, x, b, y symbolic, derive the formulas, and only then substitute $wx + b - y = -3$:

$$\frac{\partial \mathcal{L}}{\partial w} = 2(wx + b - y)x = 2(-3)(3) = -18 \quad \checkmark$$

$$\frac{\partial \mathcal{L}}{\partial b} = 2(wx + b - y) = 2(-3) = -6 \quad \checkmark$$

Backprop and the symbolic formulas agree at this point.

```
import torch

w = torch.tensor(2.0, requires_grad=True)
x = torch.tensor(3.0, requires_grad=True)
b = torch.tensor(1.0, requires_grad=True)
# Targets normally do not track gradients; enable it here to inspect dL/dy.
y = torch.tensor(10.0, requires_grad=True)

m = w * x
a = m + b
e = a - y
L = e**2
for value in (m, a, e, L):
    value.retain_grad()      # keep intermediate grads for inspection

L.backward()
print([t.item() for t in (m, a, e, L)])
# [6.0, 7.0, -3.0, 9.0]
print([t.grad.item() for t in (L, e, a, m, w, x, b, y)])
# [1.0, -6.0, -6.0, -6.0, -18.0, -12.0, -6.0, 6.0]
```

In our graph	In PyTorch	Meaning
stored scalar value	Tensor	one stored forward value
local backward rule	grad_fn	operation that produced a non-leaf tensor
saved forward value	saved by the operation	operand needed by a local derivative
reverse sweep	L.backward()	seed at $\mathcal{L}$; visit operations in reverse order
leaf derivative	p.grad	$\partial \mathcal{L} / \partial p$, accumulated with +=
intermediate derivative	retain_grad()	keep non-leaf .grad for inspection

reverse mode = direction · backprop = algorithm · autograd = engine: record → save → replay

We found $\partial \mathcal{L} / \partial w = -18$. For a small positive learning rate, which way does gradient descent move w ?

A. Decrease w

B. Increase w

C. Leave w unchanged

D. The sign tells us nothing

Answer: a negative gradient means increase the parameter

A ANSWER

Correct: B — Increase w

Why: Gradient descent subtracts the gradient: $w \leftarrow w - \eta(-18) = w + 18\eta$.

A small gradient step reduces this loss

Take $\eta = 0.01$ and update:

$$w \leftarrow 2 - 0.01(-18) = 2.18, \quad b \leftarrow 1 - 0.01(-6) = 1.06.$$

New prediction: $\hat{y} = 2.18 \cdot 3 + 1.06 = 7.60$.

loss falls from 9 to $(7.60 - 10)^2 = 5.76$

Backprop supplied the direction. The learning rate controls how far we move; a later optimization lecture studies that choice.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Build a scalar computation graph, run the forward pass, then click **backward** to watch each $\partial \mathcal{L} / \partial u$ fill in — exactly the $(wx + b - y)^2$ example above.

Trace which local rule changes when the final squared-error operation is replaced with $|e|$.

Practice the scalar graph three ways

I INTERACTIVE

See it move

Step through values and loss derivatives in the interactive graph ↗.

PyTorch → build autograd

Open the scalar-engine Colab ↓: ask PyTorch first, then rebuild this exact graph from scratch and step backward live.

Continue through the lecture

Use the complete Colab ↓ for branches, vectors, the dense VJP, batches, one update, and the tiny MLP.

Easy	change one input and recompute the forward pass
Medium	derive all four gradients before calling <code>backward()</code>
Hard	replace the square by $ e $; state what happens at $e = 0$

From scalars to vectors

We store gradients as column vectors. The name of the local object depends on the input and output shapes:

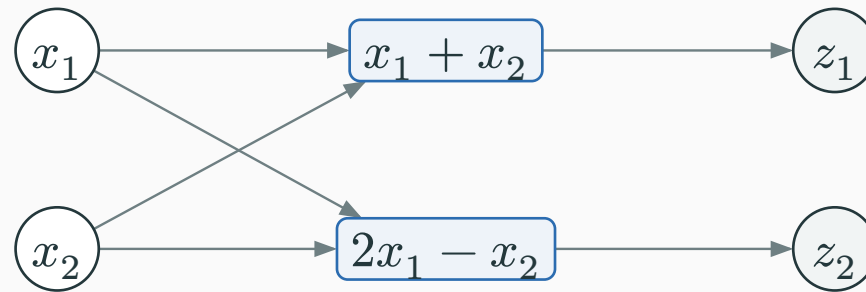
Operation shape	Local object	Name
$u \in \mathbb{R} \rightarrow v \in \mathbb{R}$	$\partial v / \partial u \in \mathbb{R}$	derivative
$\mathbf{x} \in \mathbb{R}^d \rightarrow z \in \mathbb{R}$	$\partial z / \partial \mathbf{x} \in \mathbb{R}^d$	gradient
$\mathbf{x} \in \mathbb{R}^d \rightarrow \mathbf{z} \in \mathbb{R}^m$	$J_{ij} = \partial z_i / \partial x_j$	Jacobian $\mathbf{J} \in \mathbb{R}^{m \times d}$

a Jacobian contains one local derivative for every output–input pair

A 2×2 Jacobian is four familiar derivatives

Take a vector operation with two inputs and two outputs:

$$z_1 = x_1 + x_2, \quad z_2 = 2x_1 - x_2.$$



Output	Input x_1	Input x_2
z_1	$\partial z_1 / \partial x_1 = 1$	$\partial z_1 / \partial x_2 = 1$
z_2	$\partial z_2 / \partial x_1 = 2$	$\partial z_2 / \partial x_2 = -1$

$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \cdot \text{rows are outputs} \cdot \text{columns are inputs}$$

Vector chain rule: make the shapes fit

For $x \in \mathbb{R}^d \rightarrow z \in \mathbb{R}^m$, let the **local Jacobian** be

$$J_{ij} = \partial z_i / \partial x_j, \quad J \in \mathbb{R}^{m \times d}.$$

$$\frac{\partial \mathcal{L}}{\partial x} = J^\top \frac{\partial \mathcal{L}}{\partial z}$$

$$(d \times 1) = (d \times m) (m \times 1)$$

downstream gradient = local Jacobian transpose × upstream gradient

| When $d = m = 1$, J is one number and this is the scalar chain rule from earlier.

The transpose adds every output's contribution

For the previous operation, suppose the arriving gradient is

$$\mathbf{g}_z = \partial \mathcal{L} / \partial \mathbf{z} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{J}^\top \mathbf{g}_z = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 11 \\ -1 \end{pmatrix}$$

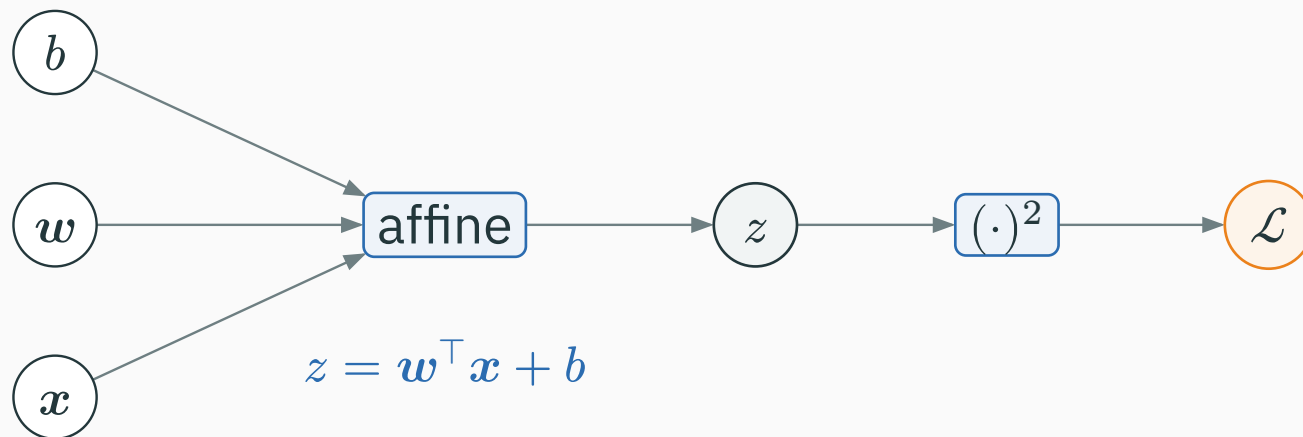
$\partial \mathcal{L} / \partial x_1$	$3(1) + 4(2) = 11$
$\partial \mathcal{L} / \partial x_2$	$3(1) + 4(-1) = -1$

the vector rule is the scalar branch rule applied to every input

An affine operation adds scalar products and a bias

Affine means a weighted sum plus a bias:

$$z = \mathbf{w}^\top \mathbf{x} + b = \sum_{j=1}^d w_j x_j + b.$$



Change one scalar while holding every other value fixed

Start from $z = \sum_j w_j x_j + b$. Change only one input by a small amount Δ :

Change	Resulting change in z	Divide by Δ
$w_i \rightarrow w_i + \Delta$	$((w_i + \Delta)x_i) - w_i x_i = x_i \Delta$	$\partial z / \partial w_i = x_i$
$x_i \rightarrow x_i + \Delta$	$w_i(x_i + \Delta) - w_i x_i = w_i \Delta$	$\partial z / \partial x_i = w_i$
$b \rightarrow b + \Delta$	$(b + \Delta) - b = \Delta$	$\partial z / \partial b = 1$

local derivative = change in the operation's output $\div$ change in one input

The derivatives with respect to the coordinates of a vector are stored in one column:

$$\begin{aligned}\frac{\partial z}{\partial \mathbf{w}} &:= \begin{pmatrix} \partial z / \partial w_1 \\ \vdots \\ \partial z / \partial w_d \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix} = \mathbf{x} \\ \frac{\partial z}{\partial \mathbf{x}} &:= \begin{pmatrix} \partial z / \partial x_1 \\ \vdots \\ \partial z / \partial x_d \end{pmatrix} = \begin{pmatrix} w_1 \\ \vdots \\ w_d \end{pmatrix} = \mathbf{w}\end{aligned}$$

With $\mathbf{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$:

$$\frac{\partial z}{\partial \mathbf{w}} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \frac{\partial z}{\partial \mathbf{x}} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \frac{\partial z}{\partial b} = 1.$$

the vector formulas are just the scalar coordinate derivatives stacked together

Affine forward: multiply componentwise, then add

Use $x = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $w = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, and $b = 0$.

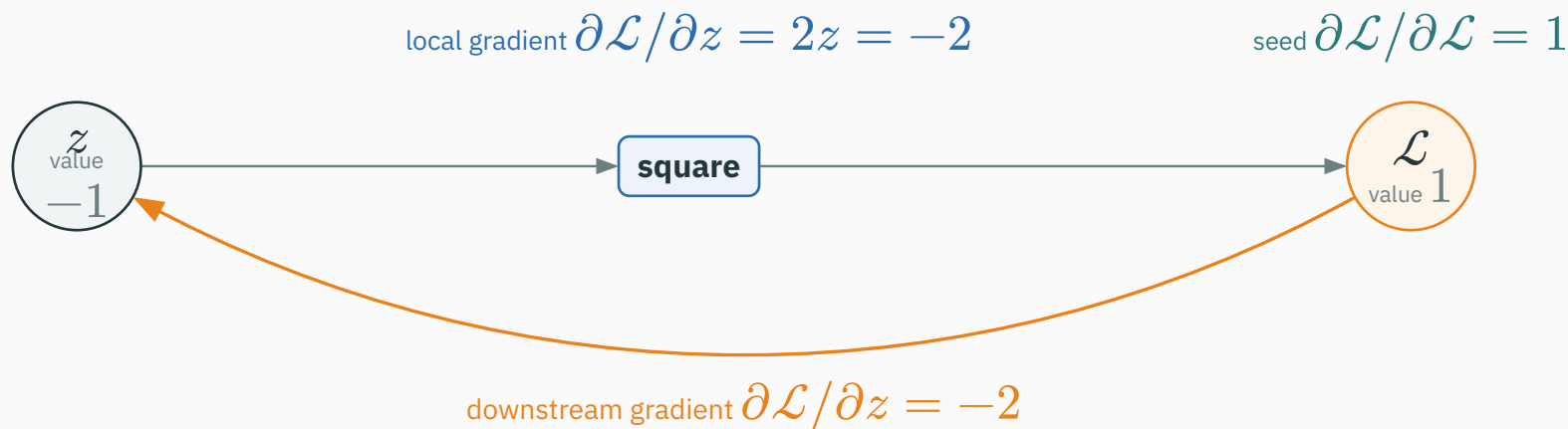
Coordinate	x_i	w_i	$w_i x_i$
1	2	1	$1 \cdot 2 = 2$
2	-1	3	$3 \cdot (-1) = -3$

$$w^\top x = 2 + (-3) = -1, \quad z = -1 + 0 = -1.$$

$$\mathcal{L} = z^2 = (-1)^2 = 1.$$

dot product = multiply matching entries, then sum

First backward step: the square operation

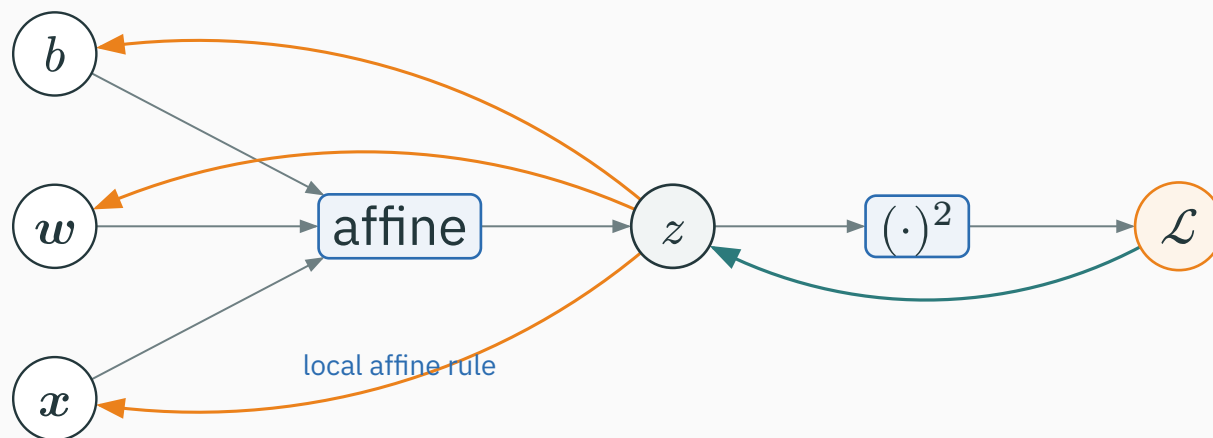


Seed $\partial \mathcal{L} / \partial \mathcal{L} = 1$. The square has local derivative $2z = -2$, so

$$\frac{\partial \mathcal{L}}{\partial z} = 1 \cdot 2z = -2.$$

the square returns -2 ; it now arrives at the affine operation

Affine backward: apply the three local gradients



Let $g_z = \partial \mathcal{L} / \partial z = -2$ arrive at the affine operation. Its **local gradients** send back:

Input	Local gradient	Downstream gradient
w	$\partial z / \partial w = x$	$\partial \mathcal{L} / \partial w = (-2)x = (-4, 2)^\top$
x	$\partial z / \partial x = w$	$\partial \mathcal{L} / \partial x = (-2)w = (-2, -6)^\top$
b	$\partial z / \partial b = 1$	$\partial \mathcal{L} / \partial b = -2$

one arriving gradient g_z produces three separate downstream gradients

PyTorch gives the same vector gradients

```
import torch

x = torch.tensor([2., -1.], requires_grad=True)
w = torch.tensor([1., 3.], requires_grad=True)
b = torch.tensor(0., requires_grad=True)

z = w @ x + b
L = z**2
L.backward()

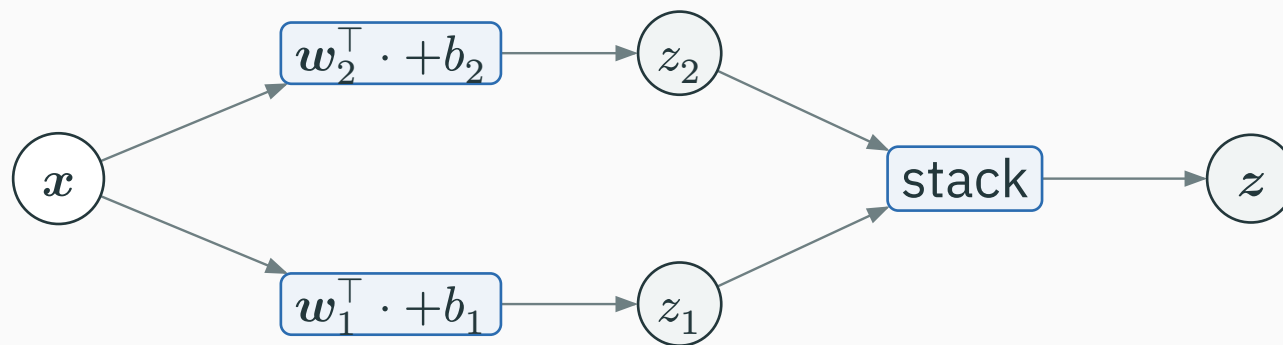
print(z.item(), L.item()) # -1.0, 1.0
print(w.grad)             # tensor([-4.,  2.])
print(x.grad, b.grad)     # tensor([-2., -6.]), tensor(-2.)
```

@ performs the dot product; backward() still applies the same local rules

From one scalar output to a vector output

A dense layer repeats one familiar neuron

Every row of $\mathbf{W}$ is one affine neuron. All rows read the same input x .



ROW 1 · neuron 1

ROW 2 · neuron 2

$$z_1 = \mathbf{w}_1^\top \mathbf{x} + b_1$$

$$z_2 = \mathbf{w}_2^\top \mathbf{x} + b_2$$

stack the outputs: $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$

Forward, row 1: do one dot product

D DERIVATION

$$\mathbf{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{w}_1^\top = (1, 3), \quad b_1 = 0$$

$$z_1 = \underbrace{1(2)}_{\text{first input}} + \underbrace{3(-1)}_{\text{second input}} + 0 = -1$$

| Nothing new happened: multiply matching entries, add them, then add the bias.

$$w_2^\top = (-2, 1), \quad b_2 = 1$$

$$z_2 = (-2)(2) + 1(-1) + 1 = -4$$

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \end{pmatrix}$$

$$W = \begin{pmatrix} 1 & 3 \\ -2 & 1 \end{pmatrix} \text{ simply stores the two neuron rows together}$$

Backward: reuse the scalar affine rule for each row

Use column vectors: $\mathbf{x} \in \mathbb{R}^{d \times 1}$, $\mathbf{W} \in \mathbb{R}^{m \times d}$, and $\mathbf{z} \in \mathbb{R}^{m \times 1}$. Suppose the later loss sends $\mathbf{g}_z = \partial \mathcal{L} / \partial \mathbf{z} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ back. For output row i ,

$$z_i = \sum_j W_{i,j} x_j + b_i, \quad g_i := \partial \mathcal{L} / \partial z_i.$$

TO x_j

TO $W_{i,j}$

TO b_i

$$\partial z_i / \partial x_j = W_{i,j}$$

contribution $g_i W_{i,j}$

$$\partial z_i / \partial W_{i,j} = x_j$$

contribution $g_i x_j$

$$\partial z_i / \partial b_i = 1$$

contribution g_i

$\mathbf{g}_z = (4, -2)^\top$ means two scalar arrivals: $g_1 = 4$ and $g_2 = -2$

The same x enters both output rows, so both rows return a contribution.

ROW 1 RETURNS

$$g_1 w_1 = 4 \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 12 \end{pmatrix}$$

ROW 2 RETURNS

$$g_2 w_2 = (-2) \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

$$g_x = \begin{pmatrix} 4 \\ 12 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 8 \\ 10 \end{pmatrix}$$

$$(g_x)_j = \sum_i g_i W_{i,j} \quad \rightarrow \quad g_x = W^\top g_z$$

Shape check: $(d \times 1) = (d \times m)(m \times 1)$. The transpose arranges the row contributions so shared input paths add.

Weight and bias gradients: separate rows stack

Output i owns row i of W , so that gradient row is $g_i x^\top$.

ROW 1

ROW 2

$$g_1 x^\top = 4(2, -1) = (8, -4)$$

$$g_2 x^\top = (-2)(2, -1) = (-4, 2)$$

$$\mathbf{g}_W = \begin{pmatrix} 8 & -4 \\ -4 & 2 \end{pmatrix} = \mathbf{g}_z \mathbf{x}^\top$$

Each bias has local derivative 1, so

$$\mathbf{g}_b = \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} = \mathbf{g}_z = \begin{pmatrix} 4 \\ -2 \end{pmatrix}.$$

returns to a shared input add · returns to separate parameter rows stack

Recipient	Coordinate rule	Stacked rule	Shape check
x	$(g_x)_j = \sum_i g_i W_{i,j}$	$g_x = W^\top g_z$	$(d \times 1) = (d \times m)(m \times 1)$
W	$(g_W)_{i,j} = g_i x_j$	$g_W = g_z x^\top$	$(m \times d) = (m \times 1)(1 \times d)$
b	$(g_b)_i = g_i$	$g_b = g_z$	$(m \times 1) = (m \times 1)$

The coordinate rules **derive** the gradients. Shape matching then verifies their orientation: $g_z x^\top$ has the same row-by-column shape as W .

From one example to a batch

A batch adds an axis—not new parameters

One example remains a column $\mathbf{x}^n \in \mathbb{R}^{d \times 1}$. A framework stores their transposes as rows of

$$\mathbf{X} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \\ 2 & 2 \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

SHARED PARAMETERS

$$\mathbf{W} \in \mathbb{R}^{m \times d}$$

$$\mathbf{b} \in \mathbb{R}^{m \times 1}$$

the same values serve every
row

BATCH FORWARD

$$\mathbf{Z} = \mathbf{X}\mathbf{W}^\top + \mathbf{1}_B\mathbf{b}^\top$$

$$(B \times m) = (B \times d)(d \times m)$$

PyTorch: `X @ W.T + b`

batching adds an example axis; every row still uses the same $\mathbf{W}, \mathbf{b}$

Three examples, three residuals

For row n , first compute $\mathbf{z}^n$, then $\mathbf{r}^n = \mathbf{z}^n - \mathbf{y}^n$.

n	$\mathbf{x}^{(n)\top}$	$\mathbf{z}^{(n)\top}$	$\mathbf{y}^{(n)\top}$	$\mathbf{r}^{(n)\top}$	ℓ_n
1	$(2, -1)$	$(-1, -4)$	$(-5, -2)$	$(4, -2)$	10
2	$(-1, 2)$	$(5, 5)$	$(7, 1)$	$(-2, 4)$	10
3	$(2, 2)$	$(8, -1)$	$(7, -2)$	$(1, 1)$	1

row 1 reproduces the earlier single example: its residual $(4, -2)^\top$ is the arriving g_z

Why the loss sends residuals backward

For one example, use half-squared error:

$$\ell_n = \frac{1}{2} \sum_k (z_k^n - y_k^n)^2.$$
$$\partial \ell_n / \partial z_k^n = z_k^n - y_k^n = r_k^n$$

THREE LOSSES

ONE SCALAR MEAN

$$\ell_1 = 10, \quad \ell_2 = 10, \quad \ell_3 = 1$$

$$\mathcal{L} = \frac{1}{3}(10 + 10 + 1) = 7$$

$$\mathbf{G}_Z = \partial \mathcal{L} / \partial \mathbf{Z} = \frac{1}{3} \mathbf{R}$$

The residual is the upstream gradient from each example's loss. The mean contributes the factor $\frac{1}{B} = \frac{1}{3}$.

One example proposes one dense-layer gradient

Reuse the single-example rules on example 1, before doing any batch algebra.

to W

to b

to x^1

$$\begin{aligned} G_W^1 &= r^1 (x^1)^\top \\ &= \begin{pmatrix} 8 & -4 \\ -4 & 2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} g_b^1 &= r^1 \\ &= \begin{pmatrix} 4 \\ -2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} g_x^1 &= W^\top r^1 \\ &= \begin{pmatrix} 8 \\ 10 \end{pmatrix} \end{aligned}$$

these are unscaled example-1 returns; its contribution to the mean loss is each value divided by 3

Each example proposes one outer product for the **same** W . Shared paths add; the mean loss scales the final sum.

EXAMPLE 1

$$G_W^1 = \begin{pmatrix} 8 & -4 \\ -4 & 2 \end{pmatrix}$$

EXAMPLE 2

$$G_W^2 = \begin{pmatrix} 2 & -4 \\ -4 & 8 \end{pmatrix}$$

EXAMPLE 3

$$G_W^3 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

$$G_W^1 + G_W^2 + G_W^3 = \begin{pmatrix} 12 & -6 \\ -6 & 12 \end{pmatrix}$$

MEAN WEIGHT GRADIENT

$$g_W = \frac{1}{3} \begin{pmatrix} 12 & -6 \\ -6 & 12 \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$$

MEAN BIAS GRADIENT

$$g_b = \frac{1}{3} \left(\begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} -2 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) =$$

Let $G_Z = \frac{R}{B} \in \mathbb{R}^{B \times m}$. The formulas below package the rowwise returns; they do not replace the derivation.

Recipient	Stacked rule	Operand shapes
W	$g_W = G_Z^\top X$	$(m \times d) = (m \times B)(B \times d)$
b	$g_b = G_Z^\top \mathbf{1}_B$	$(m \times 1) = (m \times B)(B \times 1)$
X	$g_X = G_Z W$	$(B \times d) = (B \times m)(m \times d)$

$$g_X = \frac{1}{3} \begin{pmatrix} 8 & 10 \\ -10 & -2 \\ -1 & 4 \end{pmatrix}$$

W, b are shared, so their example paths add. The three input rows are distinct, so their gradients stay in three separate rows.

Sum or mean? The reduction chooses the scale

D DERIVATION

SUM OVER EXAMPLES

$$\begin{aligned}\mathcal{L}_{\text{sum}} &= \sum_n \ell_n \\ \mathbf{G}_Z &= \mathbf{R} \\ g_W &= \mathbf{R}^\top \mathbf{X}\end{aligned}$$

MEAN OVER EXAMPLES

$$\begin{aligned}\mathcal{L}_{\text{mean}} &= \frac{1}{B} \sum_n \ell_n \\ \mathbf{G}_Z &= \frac{\mathbf{R}}{B} \\ g_W &= \mathbf{R}^\top \frac{\mathbf{X}}{B}\end{aligned}$$

```
loss = 0.5 * ((Z - Y)**2).sum(dim=1).mean()  
# sum output coordinates inside each example; mean over examples
```

the dense local rules do not change; the scalar loss reduction chooses their final scale

```
X = torch.tensor([[ 2., -1.], [-1., 2.], [2., 2.]],
                  requires_grad=True)
Y = torch.tensor([[-5., -2.], [ 7., 1.], [7., -2.]])
W = torch.tensor([[1., 3.], [-2., 1.]], requires_grad=True)
b = torch.tensor([0., 1.], requires_grad=True)

Z = X @ W.T + b
loss = 0.5 * ((Z - Y)**2).sum(dim=1).mean()
loss.backward()

print(loss.item())  # 7.0
print(W.grad)      # [[ 4., -2.], [-2.,  4.]]
print(b.grad)      # [1., 1.]
print(X.grad)      # [[ 8/3, 10/3], [-10/3, -2/3], [-1/3, 4/3]]
```

Continue in the complete lecture Colab ↗: checkpoints 11–16 derive the dense VJP, expose every batch contribution, verify microbatches, run one update, and finish with the tiny MLP.

The training loop now has a concrete batch

```
for X, Y in loader:
    optimizer.zero_grad()      # clear the previous update's buffers
    Z = model(X)               # all examples reuse the same parameters
    loss = loss_fn(Z, Y)       # reduce example losses to one scalar
    loss.backward()            # aggregate this batch's contributions
    optimizer.step()           # update once
```

<code>zero_grad()</code>	starts one intentional update window
<code>model(X)</code>	records one graph whose rows share the same parameters
<code>loss.backward()</code>	returns the reduction's sum or mean parameter gradient
<code>step()</code>	uses that gradient; it neither computes nor clears it

clear → batch forward → scalar loss → batch backward → update

One batch can arrive in pieces

Microbatches can reproduce the same mean gradient

ONE FULL BATCH

```
zero_grad()  
((ell1 + ell2 + ell3) /  
3).backward()  
step()
```

THREE PIECES · ONE UPDATE

```
zero_grad()  
(ell1 / 3).backward()  
(ell2 / 3).backward()  
(ell3 / 3).backward()  
step()
```

$$W.\text{grad} : \frac{G_W^1}{3} \rightarrow \frac{G_W^1 + G_W^2}{3} \rightarrow \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$$

Equivalence requires fixed parameters, the same overall reduction, no `step()`, and no `zero_grad()` between pieces. Batch-coupled operations such as BatchNorm need extra care.

clear once before the update window · add every scaled piece · step once after it

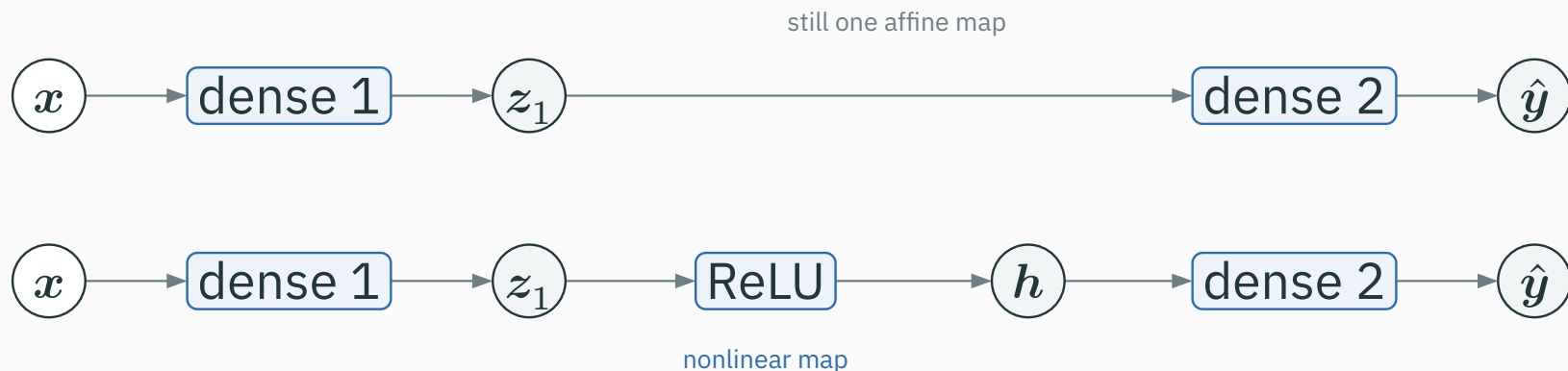
A tiny neural network

ReLU is what prevents two dense layers from collapsing into one

Without a nonlinearity, two affine maps are still one affine map:

$$\mathbf{W}_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2$$

$$= (\mathbf{W}_2\mathbf{W}_1)\mathbf{x} + (\mathbf{W}_2\mathbf{b}_1 + \mathbf{b}_2).$$



ReLU makes the composition nonlinear; depth can now represent a richer function

A tiny MLP composes three vector blocks

The network now repeats the operations we already know: affine map, elementwise ReLU, affine map, scalar loss.



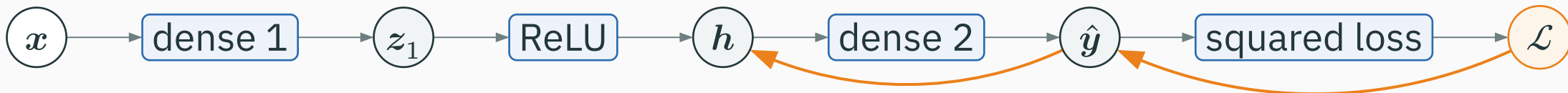
$$z_1 = W_1 x + b_1$$

$$h = \text{ReLU}(z_1)$$

$$\hat{y} = W_2 h + b_2$$

$$\mathcal{L} = \sum_i (\hat{y}_i - y_i)^2$$

dense → ReLU → dense → squared loss



1. Squared loss. Start with the seed $\partial \mathcal{L} / \partial \mathcal{L} = 1$. The loss operation returns

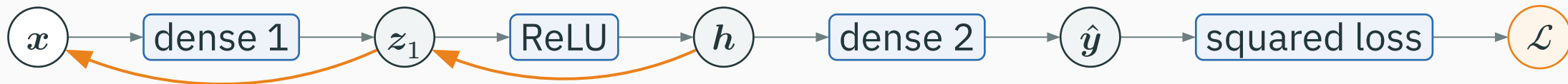
$$g_{\hat{y}} = 2(\hat{y} - y).$$

This says how increasing each prediction coordinate would change the loss. **2.**

Second dense layer. It receives $g_{\hat{y}}$ and returns a gradient to its input h :

$$g_h = W_2^\top g_{\hat{y}}.$$

the returned g_h is now the arriving gradient at ReLU



3. ReLU. Apply its local gate coordinate by coordinate:

$$g_{z_1} = g_h \odot \mathbb{1}[z_1 > 0].$$

An active unit passes its arriving gradient; an inactive unit returns zero. **4. First dense layer.** It receives g_{z_1} and returns

$$g_x = W_1^\top g_{z_1}, \quad \partial \mathcal{L} / \partial W_1 = g_{z_1} x^\top, \quad \partial \mathcal{L} / \partial b_1 = g_{z_1}.$$

each returned gradient becomes the arriving gradient for the block immediately to its left

The same tiny MLP in PyTorch

```
import torch
from torch import nn

model = nn.Sequential(
    nn.Linear(2, 3), nn.ReLU(), nn.Linear(3, 2)
)
x = torch.tensor([2., -1.])
y = torch.tensor([1., 0.])

y_hat = model(x)
loss = ((y_hat - y)**2).sum()
loss.backward()

for name, p in model.named_parameters():
    print(name, tuple(p.shape), tuple(p.grad.shape))
```

PyTorch records the three blocks and applies their local rules in reverse

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

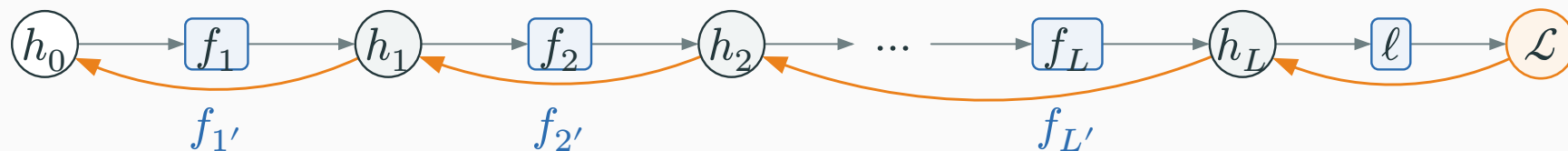
Step through the forward values, then follow one arriving gradient through a dense operation, a ReLU, and the preceding dense operation.

For practice, predict each gradient's shape before revealing its value in the interactive.

**When gradients cross many
layers**

Deep networks repeat the same multiplication

For a scalar chain $h_0 \rightarrow h_1 \rightarrow \dots \rightarrow h_L \rightarrow \mathcal{L}$:



$$\frac{\partial \mathcal{L}}{\partial h_0} = \frac{\partial \mathcal{L}}{\partial h_L} \cdot f_L'(h_{L-1}) \dots f_1'(h_0).$$

depth creates a product of many local slopes

Vector layers do the same operation with matrix–vector products. The intuition is unchanged: repeatedly small factors shrink a gradient; repeatedly large factors grow it.

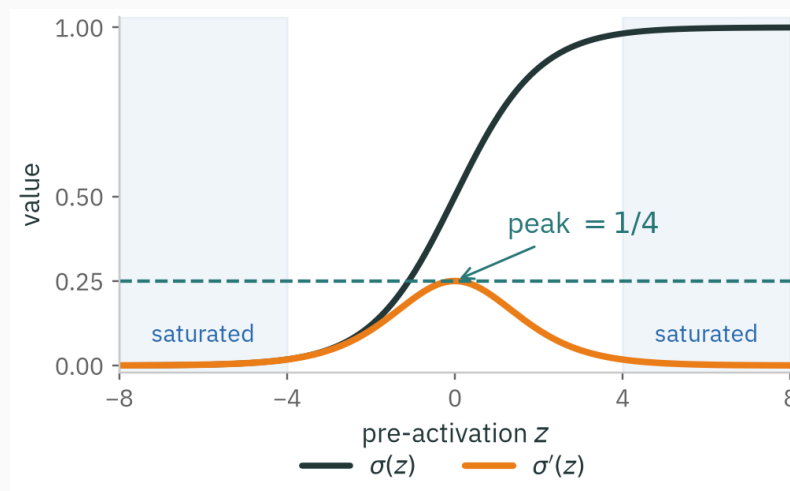
Ten ordinary-looking slopes can create three regimes

Start with an arriving loss derivative of 1 and multiply the same local slope ten times:

Local slope	After 10 layers	Effect
0.5	$0.5^{10} \approx 0.001$	gradient nearly disappears
1.0	$1^{10} = 1$	gradient is preserved
1.5	$1.5^{10} \approx 57.7$	gradient grows rapidly

the chain rule explains both vanishing and exploding gradients

$\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq \frac{1}{4}$, and ≈ 0 for large $|z|$:



each sigmoid layer multiplies the gradient by at most $1/4$

Repeated sigmoid derivatives can shrink the signal quickly. ReLU has slope 1 on active units and 0 on inactive units; weights and normalization also shape the full Jacobian.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Sweep depth and activation function; watch the gradient magnitude at layer 0 collapse for sigmoid and survive for ReLU.

Sweep depth to see repeated small local slopes rapidly erase an early-layer gradient.

Summary

Backprop in one sentence

Seed $\partial \mathcal{L} / \partial \mathcal{L} = 1$, then walk the graph **backward**, at each operation computing

downstream gradient = **upstream gradient** $\times$ **local derivative**,

and **adding** gradients wherever a variable branched.

For vector operations, the **local derivative** can be a vector or matrix; the same color-coded chain rule applies.

Setting	Backward rule
scalar operation	$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot \partial v / \partial u$
dense layer	$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}} \mathbf{x}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{W}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{z}}$
activation	$\frac{\partial \mathcal{L}}{\partial \mathbf{z}} = \frac{\partial \mathcal{L}}{\partial \mathbf{a}} \odot \varphi'(\mathbf{z})$
branch point	gradients add

We can compute $\partial \mathcal{L} / \partial \theta_j$ for every parameter.

Next: turn those gradients into reliable parameter updates.

Lecture 5 · Optimization for Deep Learning